# Flight Fare Intelligence

What drives domestic flight fares in India, and a dashboard that predicts them



Pratham Rangoonwala | Data Science Intern | Naviotech Solution Pvt. Ltd. | October 2026

# Project at a glance

**445,366**

flights, 9 airlines, 42 routes

**0.953**

$R^2$ on 89,074 held-out flights

**₹2,386**

mean absolute error per fare

**4 tabs**

live dashboard and fare predictor

## Explore

12 charts show how class, airline, route, stops, timing and booking window move fares.

## Predict

A gradient-boosted model estimates the fare of any trip, within about ₹2,400 on average.

## Deploy

A filterable Streamlit app on Streamlit Cloud, with booking tips generated from the data.

# Data and cleaning

## What is in the data

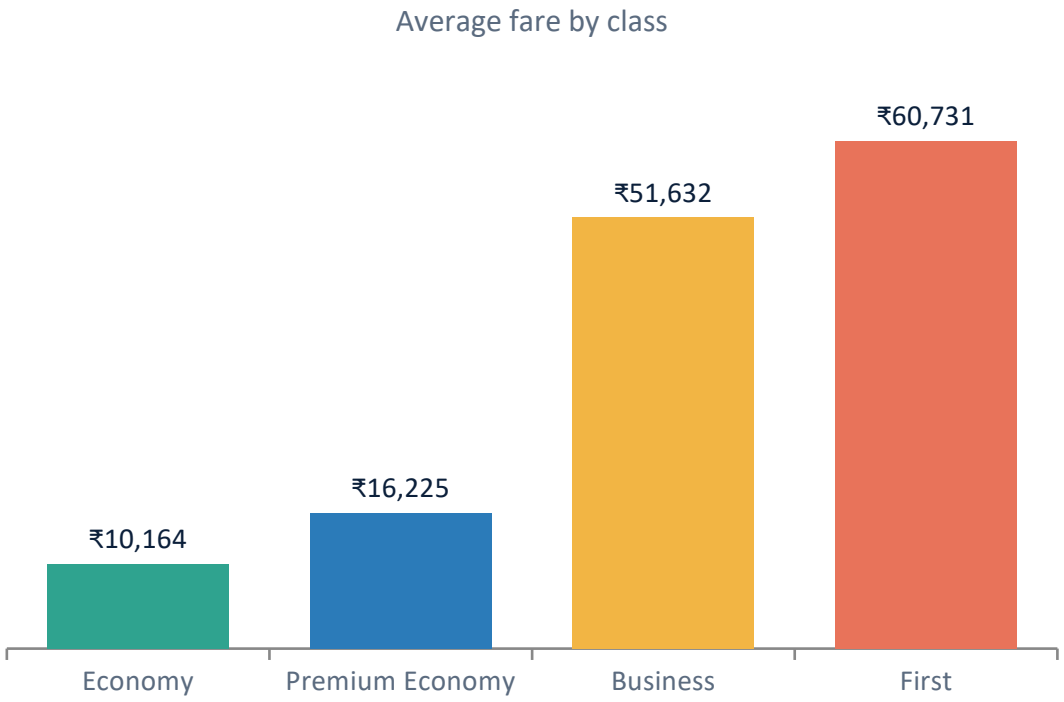
- 445,366 flights after cleaning (452,088 raw rows)
- 7 cities, 9 airlines, 42 routes
- Economy 56%, Business 28%, Premium Economy 16%; First has only 144 rows
- Departures 16 Jan to 6 Mar 2023, 1 to 50 days before takeoff

## Cleaning steps

1. Strip spaces and convert numeric columns
2. Drop missing values and 6,722 duplicate rows (1.5%)
3. Drop Date_of_journey and Flight_code
4. Convert stops to 0 / 1 / 2+ and add Route

**Limitation:** all prices were collected on a single day. Fare vs days-left patterns are real, but predictions are estimates, not live booking prices.

# Class sets the price floor




**1.6x** Premium Economy, ₹16,225 vs Economy

**5.1x** Business, ₹51,632 vs Economy

**6.0x** First, ₹60,731 (only 144 flights) vs Economy

# Last-minute flights cost more




**Fare within 3 days vs 22+ days ahead**

**+49%** Economy: ₹13,461 vs ₹9,054

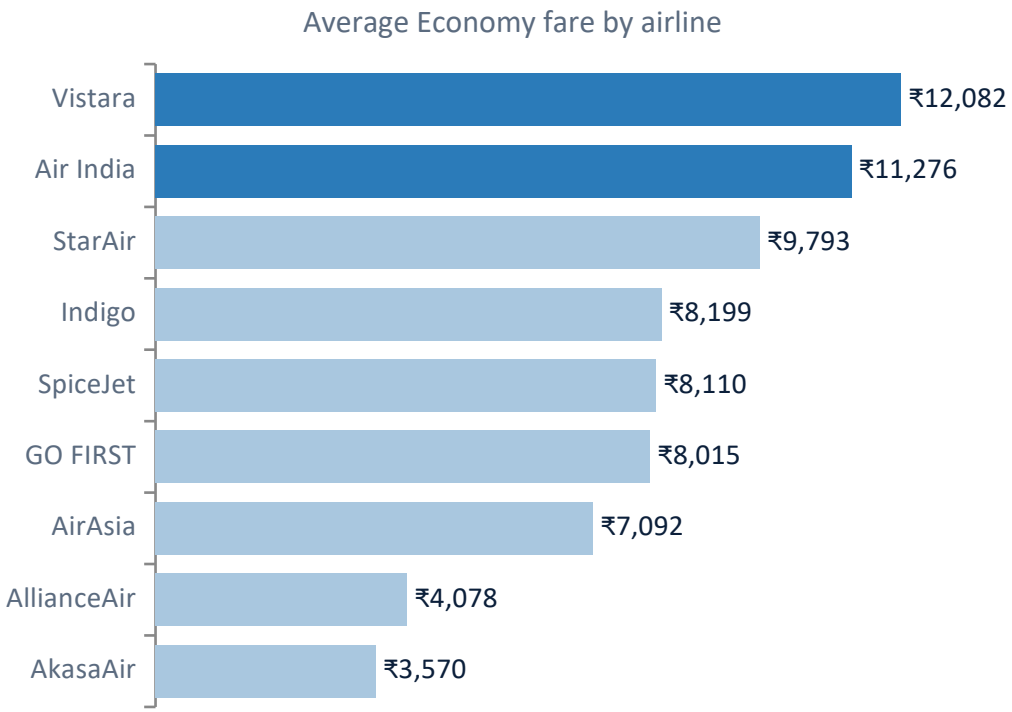
**+89%** Premium Economy

**+22%** Business

**-1%** First: 144 flights, no signal

All prices were collected on one day, so days left is entangled with calendar date. Read it as a pattern, not a law.

# Full-service airlines cost more




## Premium carriers

Vistara (₹12,082) and Air India (₹11,276) average 38 to 47% more than Indigo (₹8,199).

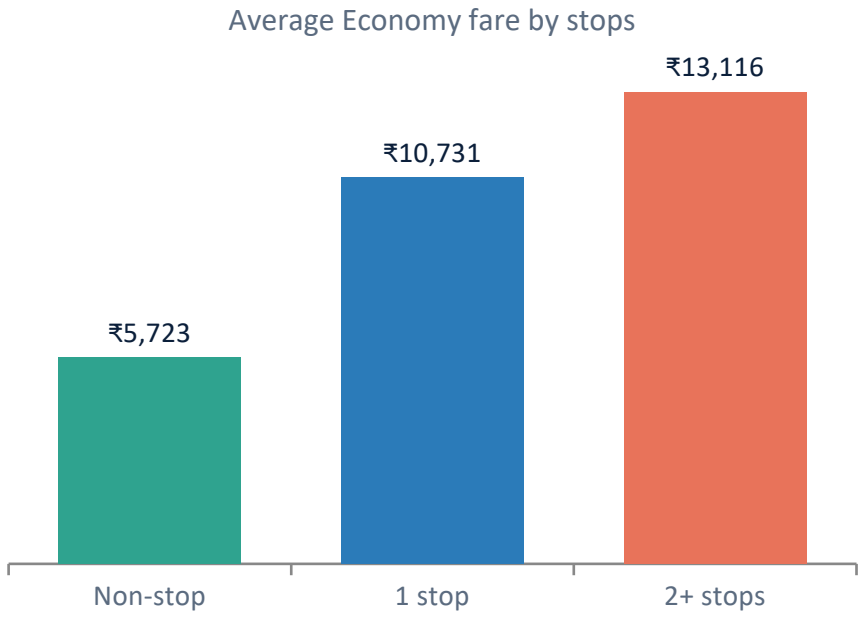
## Cheapest and steadiest

AkasaAir averages ₹3,570. AllianceAir has the steadiest fares: its middle 50% spans 31% of its median, vs 57% for Indigo.

## Mixing classes misleads

Only Air India and Vistara sell Business or First, so all-class airline averages overstate their cost.

# Stops and timing move fares




Non-stop flights are only 11.6% of all flights.

**+88%** 1 stop vs non-stop (₹10,731 vs ₹5,723), but about 12.0 h vs 2.1 h in the air

**+19%** 6 AM to 12 PM departures vs before 6 AM (₹10,696 vs ₹9,004)

**8%** gap between cheapest and costliest weekday (Tuesday ₹9,824, Sunday ₹10,640)

# Modelling approach

**10 features**
7 categorical, plus stops, duration, days left

**Encode**
OrdinalEncoder, native categorical splits

**Boost**
HistGradientBoosting on log(1 + fare)

**Ship**
joblib model, scikit-learn 1.8.0

| Model (test set) | MAE (₹) | $R^2$ |
|---|---|---|
| **Tuned Hist Gradient Boosting** | **2,386** | **0.953** |
| Random Forest | 2,357 | 0.948 |
| Decision Tree | 2,587 | 0.938 |
| Hist Gradient Boosting (untuned) | 3,120 | 0.929 |
| Linear / Ridge regression | 4,711 | 0.862 |
| Mean baseline | 16,724 | 0.000 |

## Why gradient boosting

Highest $R^2$ and lowest RMSE, MAE within about 1% of Random Forest, roughly 3x faster to train (15 s vs 41 s) and a 5.8 MB file.

Split: 80/20 stratified by class, seed 42 (356,292 train, 89,074 test). Tuning: 3-fold randomized search on 100k training rows.

# How accurate is the model?

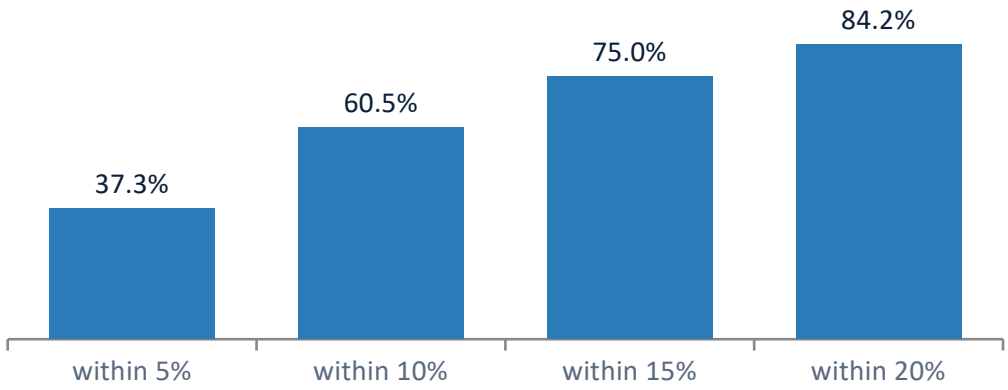
**0.953**
$R^2$ on test set

**₹2,386**
mean absolute error

**7.4%**
median error

**10.8%**
mean % error

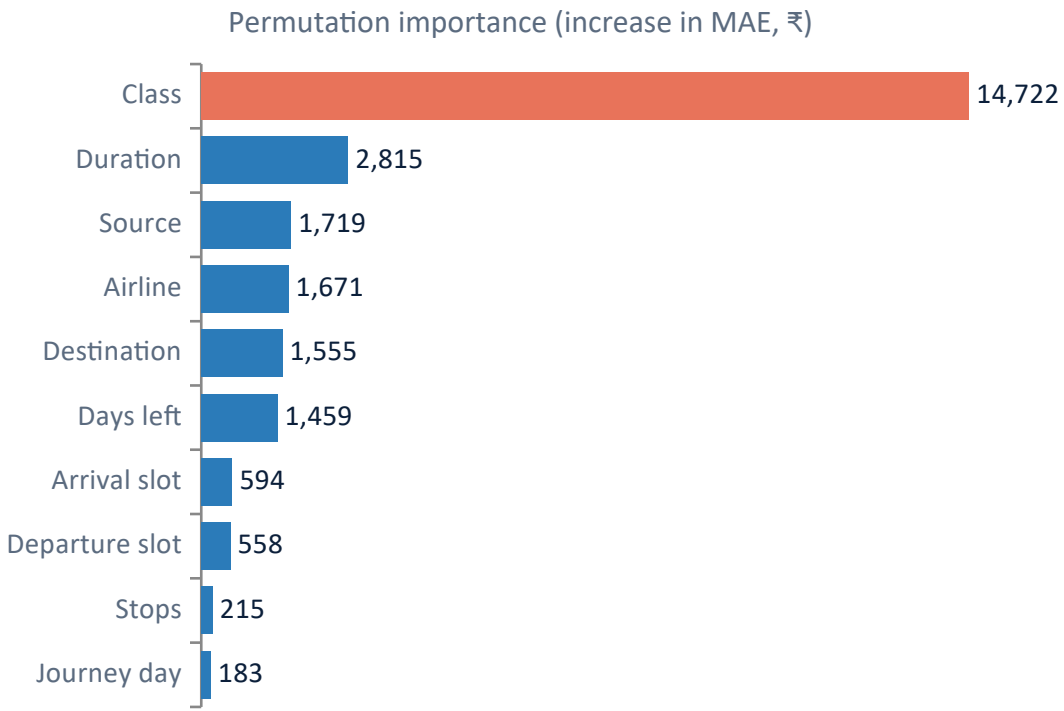
Share of test predictions within X% of the actual fare




## Where it struggles

- Error is about 9 to 12% for Economy, Premium Economy and Business, but larger in rupees on Business (₹4,927 MAE vs ₹1,213 Economy)
- Worst misses are cheap fares predicted far too high, up to about 5x

# What drives the fare




## Class dominates

Shuffling it adds ₹14,722 of error, 5x the next feature.

## Route counts twice

Source (₹1,719) and destination (₹1,555) together rival duration.

## Booking window is mid-table

Days left (₹1,459) is on par with airline (₹1,671).

## Timing is minor

Departure, arrival, stops and weekday are each under ₹600.

# The dashboard

## Overview

Class premium, days-left curve, fare distribution

## Airlines & Routes

Airline gap, spread, route heatmap, top 10, class mix

## Timing, Stops & Duration

Time of day, weekday, stops, duration vs fare

## Fare Predictor

Trip inputs to fare, range and airline comparison




12 analysis charts and 2 predictor charts, one per row, mobile-friendly.

# The fare predictor

- Eight inputs: class, route, airline, stops, days left, times, weekday, duration
- Duration auto-fills with the typical value for that airline, class, route and stops
- Shows the predicted fare against the route median and typical range
- Plots the fare across 1 to 50 days and the same trip on every airline
- Warns when the combination is not in the data



# Filters change every number

## All classes




## Class filter: Economy only




**₹22,920 to ₹10,164**

average fare

**445,366 to 248,323**

flights

**+26% to +49%**

last-minute premium

Smart Booking Tips are the exception: they use the full dataset and stay fixed.

# Engineering challenges solved

## Memory limits

Copying 445k rows on every click ran out of memory. Fix: one cached copy, a 12-column filtered frame, a 50-row pre-binned histogram and sampled box and scatter plots.

## Model version mismatch

The saved model failed with a scikit-learn _RemainderColsList error. Fix: re-saved it in Colab on the same version as the app, 1.8.0.

## Deployment error

Streamlit Cloud raised No module named '_loss'. Fix: pinned all seven packages in requirements.txt.

## Debug visibility

The sidebar shows whether the saved or the fallback model is serving. If loading fails, a matching pipeline trains itself.

# Limitations and next steps

## Limitations

- One-day price snapshot, so no true price-over-time signal
- Only about 7 weeks of departures, no festivals or holidays
- First class: 144 flights, all Air India
- Random split, so scores may overstate accuracy on future flights

## Next steps

- Collect fares across many days
- Add holidays, fuel prices and seat availability
- Quantile regression to predict fare ranges
- Time-based validation; try XGBoost and LightGBM

# Try it yourself

Live dashboard and full code, free to explore



**App:** flight-fare-intelligence.streamlit.app

**Code:** github.com/Pratham719/flight-fare-intelligence

Pratham Rangoonwala | Naviotech Solution Pvt. Ltd.